

SYNOPSIS

Report on

Finance Buddy: AI Personal Finance Manager by

Gulshan Kumari(202410116100078)

Gargi Gupta(202410116100070)

Gaurav Chauhan (202410116100073)

Gaurav Kumar(202410116100074)

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Under the supervision of

**Ms. Annu Yadav
Assistant Professor**

KIET Group of Institutions, Delhi-NCR, Ghaziabad



DEPARTMENT OF COMPUTER APPLICATIONS
**KIET GROUP OF INSTITUTIONS, DELHI-NCR,
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ABSTRACT

In the modern digital world, the use of online transactions such as UPI payments, mobile banking, and card payments has increased significantly. As a result, individuals perform numerous financial transactions every day. However, most people do not maintain proper records of their income and expenses, which often leads to overspending, poor budgeting, and lack of savings. Managing personal finances manually using notebooks or spreadsheets is time-consuming and inefficient. Therefore, there is a need for an intelligent and automated system that can help users manage their financial activities effectively.

The proposed system, **Finance Buddy**, is an AI-based personal finance management web application designed to simplify expense tracking and improve financial awareness. The system allows users to record income and expenses, categorize transactions, and monitor their financial activities in a centralized platform. It provides interactive dashboards and graphical visualizations that help users understand their spending patterns and identify areas where they can reduce expenses.

One of the key features of the system is automation. Finance Buddy integrates **Optical Character Recognition (OCR)** technology to scan bills and receipts and automatically extract transaction details, reducing manual data entry. Additionally, the system supports SMS-based transaction detection, which enables automatic expense recording by analyzing bank transaction messages. These features make the system more efficient and user-friendly.

The application also incorporates **Machine Learning techniques** to analyze historical spending data and predict future expenses. Based on these predictions, the system provides smart saving suggestions and helps users plan their monthly budgets more effectively. Furthermore, a **Financial Health Score** is generated to evaluate the overall financial condition of the user based on spending habits, savings ratio, and budget discipline.

The main objective of this project is to promote financial awareness, encourage better money management habits, and assist users in making informed financial decisions. The system is designed using modern web technologies including React, Node.js, Express, MongoDB, Python, and Machine Learning libraries.

Finance Buddy provides an integrated solution that combines expense tracking, automation, analytics, and prediction into a single platform. The proposed system has the potential to improve financial discipline and help users achieve better financial stability.

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INTRODUCTION

In the modern world, the way people manage money has changed significantly due to the rapid growth of digital technology. Online banking, debit and credit cards, mobile wallets, and UPI payments have made financial transactions faster and more convenient than ever before. Every day, individuals perform numerous transactions for daily needs such as food, transportation, shopping, entertainment, and bill payments. Although spending money has become easier, managing and tracking expenses has become more difficult for many people.

Most individuals do not maintain proper records of their income and expenses. As a result, they often lose track of where their money is being spent. This lack of financial awareness leads to overspending, poor budgeting, and reduced savings. Many people realize their financial situation only at the end of the month when they find that they have spent more than expected. This problem is especially common among students and working professionals who perform frequent digital transactions.

Traditionally, people used notebooks or spreadsheets to maintain financial records. While these methods can help in tracking expenses, they require manual effort and regular updates. Over time, users lose interest in manually recording every transaction because it is time-consuming and inconvenient. Existing personal finance applications provide basic features such as expense tracking and budgeting, but many of them still rely heavily on manual data entry and lack intelligent insights.

With the advancement of modern technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Data Analytics, it has become possible to build smarter financial management systems. These technologies can analyze historical financial data, identify spending patterns, and provide useful insights to users. Machine Learning can also predict future expenses based on past spending behavior, helping users plan their budgets more effectively. In addition, Optical Character Recognition (OCR) technology can read text from images, making it possible to automatically extract information from bills and receipts. This reduces manual work and improves the accuracy of expense tracking.

Considering these technological advancements, there is a strong need for an intelligent system that can automate the process of expense tracking and provide meaningful financial insights. To address this need, the proposed project **Finance Buddy** has been developed. **Finance Buddy** is an AI-powered personal finance management web application designed to help users track, analyze, and manage their financial activities efficiently.

The system provides a centralized platform where users can record their income and expenses, categorize transactions, and monitor their financial behavior through interactive dashboards and visual charts. One of the key features of the system is automation. Users can upload images of bills and receipts, and the system automatically extracts expense details using OCR technology. Additionally, users can paste bank SMS messages, and the system detects and records the transaction automatically. These automation features significantly reduce manual effort and make the system more user-friendly.

Another important feature of Finance Buddy is the use of Machine Learning for expense prediction. The system analyzes past spending data and predicts future expenses, helping users prepare for upcoming financial needs. The application also generates a **Financial Health Score** based on spending patterns, savings ratio, and budgeting behavior. This score provides users with a clear understanding of their financial condition and motivates them to improve their money management habits.

The main objective of this project is to promote financial awareness and encourage better financial decision-making. By combining expense tracking, automation, data analytics, and predictive modeling into a single platform, Finance Buddy aims to simplify personal finance management and help users achieve financial stability.

The system is developed using modern technologies such as React, Node.js, Express, MongoDB, Python, and Machine Learning libraries. It is designed to be secure, scalable, and easy to use, making it suitable for students, working professionals, and freelancers.

Finance Buddy provides an intelligent and automated solution for personal finance management. The proposed system has the potential to reduce financial stress, improve saving habits, and help users make informed financial decisions in their daily lives.

LITERATURE REVIEW

Personal finance management has become an important research area due to the rapid growth of digital transactions and online payment systems. With the increasing use of mobile banking, debit cards, credit cards, and UPI payments, individuals perform multiple financial transactions daily. Despite this technological advancement, many users still struggle to manage their finances effectively and fail to maintain proper records of their income and expenses.

Several existing personal finance applications such as Mint, PocketGuard, and YNAB (You Need A Budget) have been developed to help users track expenses and create budgets. These applications provide features such as transaction categorization, spending summaries, and budget planning. However, most of these systems depend heavily on manual data entry, which makes them less convenient for users. Manual entry is time-consuming and often leads to incomplete or inaccurate financial records because users may forget to record small daily expenses.

Recent research has focused on the use of Artificial Intelligence (AI) and Machine Learning (ML) in financial management systems. Machine Learning algorithms can analyze historical financial data to identify spending patterns and predict future expenses. Predictive analytics can help users plan their budgets more effectively and avoid overspending. Studies show that users who receive data-driven financial insights are more likely to improve their saving habits and maintain better financial discipline.

Another important advancement in this field is the use of Optical Character Recognition (OCR) technology. OCR allows systems to extract text from images such as bills and receipts. This technology can automate expense tracking by reducing the need for manual data entry. Research has shown that OCR-based expense management systems improve accuracy and save time for users.

In addition, data visualization techniques play a significant role in financial management applications. Graphs, charts, and dashboards help users understand their spending behavior easily. Visual analytics allow users to identify spending trends, compare income and expenses, and make informed financial decisions.

Although many applications provide individual features such as expense tracking, budgeting, or analytics, most existing systems do not offer an integrated solution that combines automation, predictive analytics, and financial health evaluation in a single platform. There is still a gap in providing a comprehensive and intelligent system that can

automate expense entry, analyze financial behavior, predict future spending, and provide personalized saving recommendations.

Therefore, the proposed system Finance Buddy aims to address these limitations by integrating expense tracking, OCR-based bill scanning, SMS transaction detection, machine learning prediction, and financial analytics into one unified platform. This integrated approach is expected to improve financial awareness and provide users with a smarter and more efficient way to manage their personal finances.

PROJECT OBJECTIVE

The primary objective of the **Finance Buddy** project is to design and develop an intelligent personal finance management system that simplifies the process of tracking, analyzing, and planning financial activities. The system focuses on improving financial awareness and helping users make better financial decisions through automation, data analytics, and machine learning.

The detailed objectives of the project are as follows:

1. To Provide a Centralized Financial Management Platform

The system aims to create a single platform where users can manage all their financial activities, including income tracking, expense recording, and budget planning. This eliminates the need for multiple tools such as notebooks, spreadsheets, or separate applications.

2. To Simplify Income and Expense Tracking

The application will allow users to record daily income and expenses in an organized manner. Transactions will be categorized into different groups such as food, travel, shopping, bills, and entertainment, making it easier to analyze spending behavior.

3. To Automate Expense Entry

One of the major objectives is to reduce manual work by automating expense entry. This will be achieved by:

- Using **OCR technology** to scan bills and receipts and extract transaction details automatically.
- Using **SMS text detection** to identify bank transaction messages and convert them into expense records.

4. To Provide Financial Data Visualization

The system will generate interactive charts and dashboards that help users understand their financial habits. Visual representations such as pie charts, bar graphs, and line charts will display spending patterns, income vs expense comparisons, and category-wise analysis.

5. To Implement Budget Planning and Monitoring

The system will allow users to set monthly budgets and monitor their spending against the set limit. If the user exceeds the budget, the system will generate alerts to help control unnecessary expenses.

6. To Predict Future Expenses Using Machine Learning

The project aims to use machine learning algorithms to analyze historical financial data and predict future expenses. This prediction will help users plan their finances more effectively and avoid unexpected financial shortages.

7. To Generate a Financial Health Score

The system will calculate a financial health score based on savings ratio, spending habits, and budgeting discipline. This score will provide users with an overview of their financial condition and motivate them to improve their money management skills.

8. To Provide Smart Saving Suggestions

Based on user spending patterns, the system will provide personalized saving tips and recommendations to help users reduce unnecessary expenses and improve savings.

9. To Promote Financial Awareness

The overall goal of the project is to encourage financial discipline and help users make informed financial decisions using data-driven insights.

HARDWARE AND SOFTWARE REQUIREMENTS

1. Hardware Requirements

The Finance Buddy system is a web-based application and does not require any specialized hardware. It can be developed and executed on a standard computer system.

For Development:

- Processor: Intel Core i3 / AMD Ryzen 3 or higher (i5 recommended)
- RAM: Minimum 4 GB (8 GB recommended for smooth performance)
- Storage: At least 20 GB free disk space (SSD preferred)
- Display: Minimum resolution 1366×768
- Internet Connection: Required for development, testing, and deployment
- Input Devices: Keyboard and Mouse

For End Users:

- Laptop, Desktop, Tablet, or Smartphone
- Modern web browser
- Stable internet connection

Optional Hardware:

- Smartphone camera (for capturing bill images for OCR scanning)

2. Software Requirements

The system is developed using modern web and AI technologies.

Frontend Technologies:

- HTML
- CSS
- JavaScript

- React.js

Backend Technologies:

- Node.js
- Express.js

Database:

- MongoDB

Artificial Intelligence / Machine Learning:

- Python
- Pandas
- Scikit-learn

OCR Technology:

- Tesseract OCR

Development Tools:

- Visual Studio Code
- Git & GitHub
- Postman (API testing)

Web Browser:

- Google Chrome / Microsoft Edge / Mozilla Firefox

PROJECT METHODOLOGY

The development of **Finance Buddy** follows a structured approach to ensure proper planning, development, testing, and deployment of the system. The **Agile methodology** is adopted for this project, as it allows continuous development, testing, and improvement in different phases.

1. Requirement Analysis

In this phase, the problem of personal finance management was studied and the requirements of the system were identified.

The following key requirements were defined:

- Expense tracking and categorization
- Budget planning and alerts
- Automation using OCR and SMS detection
- Machine learning based expense prediction
- Financial analytics dashboard

2. System Design

After gathering requirements, the system architecture and design were created.

This phase included:

- Designing system architecture
- Database design (MongoDB collections)
- Creating system diagrams (DFD, ER Diagram, Use Case)
- Planning user interface and workflow

3. Development Phase

The development phase was divided into frontend and backend development.

Frontend Development

- User interface developed using React.js
- Login, dashboard, and expense pages created

Backend Development

- APIs developed using Node.js and Express.js
- Database integration using MongoDB

4. Integration of Advanced Features

After core development, advanced features were added:

- OCR integration for bill scanning
- SMS transaction detection
- Machine Learning model for expense prediction
- Financial health score generation

5. Testing Phase

The system was tested to ensure reliability and performance.

Testing included:

- Functional testing
- Integration testing
- User interface testing
- Bug fixing and performance improvement

6. Deployment

Finally, the application was prepared for deployment and documentation was completed.

This methodology ensured systematic development and successful completion of the project.

PROJECT OUTCOME

The Finance Buddy project was successfully developed as an AI-based personal finance management system that helps users track, analyze, and manage their financial activities efficiently.

The system provides a centralized platform where users can record income and expenses, categorize transactions, and monitor their spending behavior through interactive dashboards and graphical charts. The application successfully reduces manual effort by introducing automation features such as bill scanning using OCR and SMS-based transaction detection.

The integration of Machine Learning enables the system to analyze historical financial data and predict future expenses. Based on this analysis, the system provides smart saving suggestions and generates a financial health score that helps users understand their financial condition.

The project demonstrates how modern web technologies and AI can be combined to create a practical and user-friendly solution for personal finance management. The developed system improves financial awareness, encourages better budgeting habits, and supports users in making informed financial decisions.

Overall, Finance Buddy achieves its goal of simplifying expense tracking, promoting financial discipline, and providing intelligent insights for better money management.

PROPOSED TIME DURATION

Phase	Activity	Duration
Phase 1	Project Planning & Topic Finalization	Week 1
Phase 2	Requirement Analysis & SRS Preparation	Week 2 – 3
Phase 3	System Design & Diagrams	Week 4 – 5
Phase 4	Frontend Development	Week 6 – 8
Phase 5	Backend Development & Database Integration	Week 9 – 11
Phase 6	OCR & SMS Automation Integration	Week 12 – 13
Phase 7	Machine Learning Model Development	Week 14 – 16
Phase 8	Testing & Debugging	Week 17 – 19
Phase 9	Deployment	Week 20

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